

HYDROGENATED NBR SYNTHESIS AND PROCESS LANDSCAPE FROM SELECTED PATENTS

1. ABSTRACT

This report synthesizes patent evidence relevant to hydrogenated nitrile butadiene rubber (HNBR), focusing on how NBR precursors are converted to partially or highly hydrogenated elastomers with reduced residual unsaturation, improved heat/oil resistance, and improved processability. The selected patent landscape spans two major technical directions: direct hydrogenation of NBR in solution or specialized reactors, and coupled metathesis/hydrogenation routes used to lower molecular weight before or during hydrogenation. Across the set, the target material is consistently defined as a nitrile rubber copolymer or terpolymer containing at least one conjugated diene and at least one α,β -unsaturated nitrile, optionally with additional comonomers; hydrogenation is used to convert some or all diene-derived C=C bonds to saturated backbone units while preserving nitrile functionality.

The patents show several process families. One group emphasizes reactor/process intensification, including spinning disk reactor hydrogenation with hydrogen pressures from 3.5 to 100 bar and homogeneous catalysts based on Rh/Ru and related metals. Another group focuses on simultaneous metathesis and hydrogenation using ruthenium/osmium catalyst systems in the presence of hydrogen and optional co-olefins, enabling single-step reduction of molecular weight and residual double bond content. A third group uses metathesis first, then hydrogenation, to obtain low-MW HNBR with narrow molecular weight distributions. Additional patents address upstream NBR synthesis in organic solvents or solvent mixtures, including emulsion and solution polymerization with mercaptan chain transfer control, because precursor molecular weight strongly influences final HNBR viscosity. Finally, one patent addresses post-hydrogenation purification by removing ruthenium residues with functionalized ion-exchange resins, and two patents apply partially hydrogenated NBR in carbon nanotube dispersions, where residual double bond content and monomer composition are tuned for dispersion performance.

Overall, the evidence indicates that the most relevant synthesis levers for HNBR are controlled NBR precursor composition, selective hydrogenation conditions, catalyst choice, hydrogen pressure, and, where needed, metathesis-assisted molecular weight reduction. The strongest process trend is toward lower-viscosity, lower-residual-unsaturation HNBR made either by intensified hydrogenation or by one-step/ two-step metathesis-hydrogenation sequences.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses catalyst systems for metathesis of nitrile rubber, using a metathesis catalyst together with one or more salts to accelerate degradation of NBR. The process is aimed at reducing molecular weight before subsequent hydrogenation to HNBR, rather than hydrogenating directly in the same step. - Salt(s): one or more salts of general formula $K_n^+ A_z^-$ — claim 1 - Salt:metathesis catalyst weight ratio: 0.01:1 to 10000:1 — claim 1 - Example 7: 10 mg Grubbs II catalyst — Example 7 - Example 7: 1 mg LiBr — Example 7.02 - Example 7: 0.3 ml chlorobenzene — Example 7 - Example 7: 0.2 ml $CDCl_3$ — Example 7 - Example 7: 0.15 ml diethyl diallylmalonate — Example 7 - Example 7: room temperature — Example 7 - Example 8: 1 mg CsBr — Example 8.02 - Example 8: 0.3 ml chlorobenzene — Example 8 - Example 8: 0.2 ml $CDCl_3$ — Example 8 - Example 8: 0.15 ml diethyl diallylmalonate — Example 8 - Example 8: room temperature — Example 8

Relevant Experimental Evidence - Example 7: LiBr addition accelerated ring-closing metathesis relative to no salt. - Example 8: CsBr addition was tested analogously to LiBr and showed a salt effect on metathesis conversion.

Technical Relevance Although not a hydrogenation patent per se, it is relevant because it teaches metathesis catalyst/salt systems for controlled NBR chain scission, a precursor strategy used in low-MW HNBR manufacture. The disclosed salt-accelerated metathesis concept informs upstream molecular-weight reduction before hydrogenation.

Patent 2

- Patent Number: US7666950B2
- Patent Title: Process for preparing hydrogenated nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparing HNBR by hydrogenating NBR in a spinning disk reactor. The process is designed to lower hydrogenation severity and equipment cost, and it may use homogeneous hydrogenation catalysts, especially Rh- or Ru-based systems. - Hydrogen pressure: 3.5 bar to 100 bar — claim 1 - Hydrogen pressure: 65 bar to 90 bar — background discussion of batch hydrogenation - NBR composition: 40 to 90 wt.% conjugated diene — detailed description - NBR composition: 10 to 60 wt.% α,β -unsaturated nitrile — detailed description - Additional comonomers: 0 to 40 wt.% — detailed description - NBR Mooney viscosity: 30 to 90 ML 1+4 @100°C — detailed description - NBR Mooney viscosity: 30 to 70 ML 1+4 @100°C — detailed description - NBR Mooney viscosity: 30 to 50 ML 1+4 @100°C — detailed description - NBR Mw: 200,000 to 500,000 — detailed description - NBR Mn: 80,000 to 150,000 — detailed description - NBR PDI: 2.0 to 6.0 — detailed description - HNBR Mooney viscosity: 55 to 120 ML 1+4 @100°C — background discussion - HNBR Mw: 200,000 to 500,000 — background discussion - HNBR residual double bond content: up to 18% — background discussion - Low-Mooney HNBR: 2 to 50 ML 1+4 @100°C — background discussion - Low-Mooney HNBR Mw: 30,000 to 250,000 — background discussion - Catalyst classes: rhodium, ruthenium, titanium, platinum, iridium, palladium, rhenium, osmium, cobalt or copper — claim 5 - Catalyst formula: $(R_1mB)IMX_n$ — claim 7

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This is a core HNBR process patent because it directly addresses hydrogenation of NBR to HNBR and specifies reactor and pressure windows. It is especially relevant for process intensification and for reducing the high-pressure burden associated with conventional batch hydrogenation.

Patent 3

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low molecular weight HNBR prepared by first reducing NBR molecular weight through olefin metathesis and then hydrogenating the degraded polymer. The process is aimed at producing HNBR with lower Mooney viscosity, narrower molecular weight distribution, and improved processability. - HNBR MW: 30,000 to 250,000 g/mol — claim 1 - HNBR Mooney viscosity: less than 10 ML 1+4 @100°C — claim 1 - HNBR MWD/PDI: less than 2.5 — claim 1 - HNBR MW: 40,000 to 220,000 g/mol —

claim 2 - HNBR PDI: less than 2.3 — claim 3 - Example 1: 200 g rubber — Example 1 - Example 1: 1133 g MCB — Example 1 - Example 1: 15 wt.-% solid — Example 1 - Example 1: degassed 3 times with C₂H₄ — Example 1 - Example 1: 6.9×10⁵ Pa — Example 1 - Example 2: 200 g rubber — Example 2 - Example 2: 1133 g MCB — Example 2 - Example 2: 15 wt.-% solid — Example 2 - Example 2: degassed 3 times with C₂H₄ — Example 2 - Example 2: 6.9×10⁵ Pa — Example 2 - Example 3: 75 g rubber — Example 3 - Example 3: 1175 g MCB — Example 3 - Example 3: 6 wt.-% solid — Example 3 - Example 3: degassed 3 times with C₂H₄ — Example 3 - Example 3: 6.9×10⁵ Pa — Example 3 - Example 4: 75 g rubber — Example 4 - Example 4: 1175 g MCB — Example 4 - Example 4: 6 wt.-% solid — Example 4 - Example 4: 150 g 1-hexene — Example 4 - Example 4: degassed 3 times with dry N₂ — Example 4 - Example 4: 80°C reactor temperature — Table 1 - Example 4: 0.05 phr catalyst load — Table 1 - Example 4: 0.10 phr catalyst load — Table 1 - Example 4: 0.25 phr catalyst load — Table 1 - Example 4: 500 psi co-olefin concentration — Table 1 - Example 4: Mn 27 kg/mol — Table 2 - Example 4: Mw 54 kg/mol — Table 2 - Example 4: PDI 2.0 — Table 2 - Example 4: Mooney viscosity 3 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis of NBR cement was demonstrated at 6.9×10⁵ Pa. - Example 2: A second ethylene metathesis run confirmed the same solvent/pressure setup. - Example 3: Lower-solids cement still underwent metathesis under ethylene pressure. - Example 4: 1-Hexene was used as co-olefin and gave low-MW product with Mw 54 kg/mol and Mooney 3. - Example 4: Catalyst loading and 80°C conditions were compared in Table 1.

Technical Relevance This is one of the most directly relevant patents because it couples NBR metathesis with hydrogenation to produce low-MW HNBR. It provides concrete evidence that molecular-weight reduction before hydrogenation can dramatically lower Mooney viscosity and narrow PDI.

Patent 4

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent also discloses low molecular weight HNBR made by metathesis of NBR followed by hydrogenation. It emphasizes the same general route as the related low-MW HNBR patent family, with examples showing solvent-based metathesis using ethylene or 1-hexene and subsequent property reduction. - HNBR MW: 30,000 to 250,000 g/mol — claim 1 - HNBR Mooney viscosity: 3 to 50 ML 1+4@100°C — claim 1 - HNBR MWD/PDI: less than 2.5 — claim 1 - Conjugated diene content: 40 to 85 wt.% —

claim 5 - Unsaturated nitrile content: 14 to 50 wt.% — claim 5 - Example 1: 200 g rubber — Example 1 - Example 1: 1133 g MCB — Example 1 - Example 1: 15 wt.-% solid — Example 1 - Example 1: C₂H₄ at 6.9×10⁵ Pa — Example 1 - Example 2: 200 g rubber — Example 2 - Example 2: 1133 g MCB — Example 2 - Example 2: 15 wt.-% solid — Example 2 - Example 2: C₂H₄ at 6.9×10⁵ Pa — Example 2 - Example 3: 75 g rubber — Example 3 - Example 3: 1175 g MCB — Example 3 - Example 3: 6 wt.-% solid — Example 3 - Example 3: C₂H₄ at 6.9×10⁵ Pa — Example 3 - Example 4: 75 g rubber — Example 4 - Example 4: 1175 g MCB — Example 4 - Example 4: 6 wt.-% solid — Example 4 - Example 4: 150 g 1-hexene — Example 4 - Example 4: dry N₂ degassing — Example 4 - Example 4: 80°C reactor temperature — Table 1 - Example 4: 0.05 phr catalyst load — Table 1 - Example 4: 0.10 phr catalyst load — Table 1 - Example 4: 0.25 phr catalyst load — Table 1 - Example 4: 500 psi co-olefin concentration — Table 1 - Example 4: Mn 27 kg/mol — Table 2 - Example 4: Mw 54 kg/mol — Table 2 - Example 4: PDI 2.0 — Table 2 - Example 4: Mooney viscosity 3 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene metathesis at 15 wt.-% solids was demonstrated. - Example 2: Ethylene metathesis was repeated under the same solvent and pressure conditions. - Example 3: A lower-solids cement still gave strong molecular-weight reduction. - Example 4: 1-Hexene co-olefin produced a low-MW product with Mw 54 kg/mol and Mooney 3. - Example 4: The table shows catalyst loading and 80°C reactor temperature effects.

Technical Relevance This patent is highly relevant because it defines the low-MW HNBR target space and shows how metathesis-assisted chain scission can be used before hydrogenation. The disclosed composition and process windows are useful for designing feedstocks that hydrogenate to low-viscosity HNBR.

Patent 5

- Patent Number: EP2554558A1
- Patent Title: Method for producing nitrile rubbers in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparing NBR in organic solvent mixtures by free-radical polymerization, with the resulting nitrile rubber then optionally hydrogenated by known methods. The emphasis is on solvent selection and polymerization control to improve time-conversion behavior and produce NBR suitable for later HNBR conversion. - Main solvent: 70 to 99.9 vol% of total solvent — claim 1 - Example 1: 273.7 mg Vam 110 — Example 1 - Example 1: 0.88 mmol Vam 110 — Example 1 - Example 1: 0.38 phm Vam 110 — Example 1 - Example 1: 61.4 mg DoPAT — Example 1 - Example 1: 0.175 mmol DoPAT — Example 1 - Example 1: 0.086 phm DoPAT — Example 1 - Example 1: 90 ml

monochlorobenzene — Example 1 - Example 1: 139 phm monochlorobenzene — Example 1 - Example 1: 5 ml dimethylacetamide — Example 1 - Example 1: 6.6 phm dimethylacetamide — Example 1 - Example 1: 31.7 ml acrylonitrile — Example 1 - Example 1: 481.5 mmol acrylonitrile — Example 1 - Example 1: 36 phm acrylonitrile — Example 1 - Example 1: 10 min nitrogen degassing — Example 1 - Example 1: 70.5 ml 1,3-butadiene — Example 1 - Example 1: 847.2 mmol 1,3-butadiene — Example 1 - Example 1: 64 phm 1,3-butadiene — Example 1 - Example 1: 100°C polymerization temperature — Example 1 - Example 1: 22 h reaction time — Example 1

Relevant Experimental Evidence - Example 1: NBR was prepared from acrylonitrile and 1,3-butadiene in monochlorobenzene/dimethylacetamide. - Example 1: Polymerization was initiated at 100°C after nitrogen degassing. - Example 1: The example demonstrates solvent-mixture control and a 22 h polymerization window.

Technical Relevance This patent is relevant as an upstream precursor process: HNBR quality depends strongly on the NBR feedstock molecular weight and composition. The solvent-mixture polymerization route can be used to tailor NBR before hydrogenation.

Patent 6

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous metathesis and hydrogenation of nitrile rubber in the presence of hydrogen, using ruthenium or osmium catalyst systems. The process is designed to produce low molecular weight HNBR with low residual double bond content in a single step. - Molecular weight Mw: 20,000 to 250,000 — description - Mooney viscosity: 1 to 50 ML 1+4 @100°C — description - MWD/PDI: less than 3.0 — description - NBR Mooney viscosity: 24 to 70 ML 1+4 at 100°C — description - NBR Mooney viscosity: 28 to 40 ML 1+4 at 100°C — description - NBR Mw: 200,000 to 400,000 — description - NBR Mw: 200,000 to 500,000 — description - NBR Mn: 80,000 to 150,000 — description - NBR PDI: 2.0 to 6.0 — description - Conjugated diene: C4-C6 — description - 1,3-butadiene: especially preferred — description - Isoprene: preferred — description - 2,3-dimethylbutadiene: preferred — description - Piperylene: preferred — description - α,β -unsaturated nitrile: C3-C5 — description - Acrylonitrile: particularly preferred — description - Methacrylonitrile: preferred — description - Ethacrylonitrile: preferred — description - Additional comonomers: 0 to 40 wt.% — description - Additional comonomers: 0.1 to 40 wt.% — description - Additional comonomers: 1 to 30 wt.% — description - Hydrogenation catalyst: compound of general formula (I) — claim 1 - M is ruthenium or osmium — claim 1 - Hydrogen present: yes — claim 1 - Co-olefin: optional — claim 1 - Example

1: Hoveyda catalyst loading 0.041 phr — Example 1 - Example 1: Arlt catalyst loading 0.045 phr — Example 1 - Example 1: Wilkinson's catalyst loading 0.060 phr — Example 1 - Example 1: metal content 65.6 ppm — Example 1 - Example 1: triphenylphosphine co-catalyst 1.0 phr — Example 1 - Example 1: reaction time 0 to 240 min — Example 1 - Example 1: RDB 100% to 0.4% — Example 1 - Example 1: Mw 113371 g/mol — Example 1 - Example 1: Mn 48352 g/mol — Example 1 - Example 1: PDI 2.34 — Example 1 - Example 1: Mooney viscosity 33.3 — Example 1 - Example 1: Mw 133758 g/mol — Example 1 - Example 1: Mn 54595 g/mol — Example 1 - Example 1: PDI 2.45 — Example 1 - Example 1: Mooney viscosity 41.3 — Example 1 - Example 1: Mw 195304 g/mol — Example 1 - Example 1: Mn 64130 g/mol — Example 1 - Example 1: PDI 3.05 — Example 1 - Example 1: Mooney viscosity 69.3 — Example 1

Relevant Experimental Evidence - Example 1: Three catalyst systems were compared for simultaneous hydrogenation/metathesis. - Example 1: Hoveyda catalyst gave the fastest RDB reduction to 0.4% by 240 min. - Example 1: The inventive catalyst reduced Mw to 113,371 g/mol and Mooney to 33.3. - Example 1: Wilkinson's catalyst with triphenylphosphine gave higher residual RDB and higher Mooney. - Example 1: The data show that catalyst choice strongly affects both hydrogenation efficiency and molecular-weight reduction.

Technical Relevance This is one of the most important patents for HNBR synthesis because it directly teaches single-step simultaneous hydrogenation and metathesis. It provides a route to low-MW, low-RDB HNBR and identifies catalyst systems that outperform the industrial benchmark.

Patent 7

- Patent Number: US7846995B2
- Patent Title: ... -containing catalyst residues from optionally hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses removal of ruthenium-containing catalyst residues from optionally hydrogenated nitrile rubber using functionalized ion-exchange resins. The process is a downstream purification step for metathesis/hydrogenation-derived HNBR, aimed at lowering residual Ru content in the polymer solution and recovered rubber. - Ruthenium in solution: 5 to 1000 ppm — claim 2 - Ruthenium in solution: 5 to 500 ppm — claim 3 - Rubber concentration in solution: 0.5 to 30 wt.% — claim 4 - Ion exchange resin particle diameter: 0.2 to 2.5 mm dry basis — claim 1 - Example 2A: 0.5 g Lewatit MonoPlus MP 500 — Example 2A - Example 2A: 180 g standard nitrile rubber solution — Example 2A - Example 2A: ca. 100°C — Example 2A - Example 2A: 66 h — Example 2A - Example 2A: 60°C reduced pressure oven drying — Example 2A - Example 2A: 10 ppm Ru after treatment — Example 2A - Example 2A: 33% Ru removal — Example 2A

Relevant Experimental Evidence - Example 2A: Functionalized ion-exchange resin reduced Ru content in rubber solution to 10 ppm. - Example 2A: The process achieved 33% Ru removal after 66 h at about 100°C. - Example 2B: Untreated control rubber was used to benchmark residual Ru content.

Technical Relevance This patent is relevant because catalyst residue removal is critical for HNBR made by ruthenium-catalyzed metathesis/hydrogenation. It provides a practical purification route to low-metal HNBR suitable for demanding applications.

Patent 8

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance Selected via deterministic pipeline scoring.

Patent 9

- Patent Number: US10686185B2
- Patent Title: Carbon nanotube dispersion liquid and manufacturing method thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses carbon nanotube dispersion liquids using partially hydrogenated nitrile rubber as the dispersant/binder. The partially hydrogenated NBR is prepared by copolymerizing α,β -unsaturated nitrile and conjugated diene monomers and then hydrogenating the C=C bonds to a controlled residual double bond level. - RDB value: 0.5 to 40 wt.% — claim 1 - D50 particle size: 2 to 5 μm — claim 1 - α,β -unsaturated nitrile-derived unit: 20 to 50 wt.% — claim 3 - Mooney viscosity: 10 to 120 ML 1+4 at 100°C — description - Mooney viscosity: 10 to 100 ML 1+4 at 100°C — description - Dispersion medium

content: 85 to 99 wt.% — description - Whole solute content: 1 to 15 wt.% — description - Carbon nanotube content: 50 to 90 wt.% of solute — description - Example 7: D10 1.06 μm — Example 7 - Example 7: D50 4.08 μm — Example 7 - Example 7: D90 9.15 μm — Example 7 - Example 7: viscosity 34.0 Pa·s — Example 7 - Example 7: dispersion efficiency 37.8 kWh/kg — Example 7 - Example 7: adhesion 31 gf/cm — Example 7 - Example 7: DC-IR 1.236 ohm — Example 7 - Example 8: D10 1.15 μm — Example 8 - Example 8: D50 2.35 μm — Example 8 - Example 8: D90 4.98 μm — Example 8 - Example 8: viscosity 52 Pa·s — Example 8 - Example 8: dispersion efficiency 35.7 kWh/kg — Example 8 - Example 8: adhesion 28 gf/cm — Example 8 - Example 8: DC-IR 1.252 ohm — Example 8

Relevant Experimental Evidence - Example 7: Bundle-type CNTs dispersed with partially hydrogenated NBR gave D50 4.08 μm and good battery/electrode metrics. - Example 8: A second formulation showed D50 2.35 μm and maintained dispersion performance. - Examples 1-6: Comparative formulations illustrate the importance of the partially hydrogenated NBR dispersant for CNT wetting.

Technical Relevance This patent is indirectly relevant to HNBR synthesis because it defines useful partially hydrogenated NBR composition windows and residual unsaturation levels. It shows how hydrogenation degree and monomer composition affect solubility, viscosity, and interfacial performance in advanced formulations.

Patent 10

- Patent Number: EP3348582B1
- Patent Title: Carbon nanotube dispersion liquid and manufacturing method thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses carbon nanotube dispersion liquids using partially hydrogenated nitrile rubber with controlled residual double bond content. The partially hydrogenated NBR is prepared by copolymerizing nitrile and diene monomers and then hydrogenating the backbone, with the resulting material used to wet and disperse CNTs in organic media. - RDB value: 0.5 to 40 wt.% — claim 1 - D50 particle size: 3 to 10 μm — claim 1 - α,β -unsaturated nitrile-derived unit: 20 to 50 wt.% — claim 2/3 - Mooney viscosity: 10 to 120 ML 1+4 at 100°C — description - Mooney viscosity: 10 to 100 ML 1+4 at 100°C — description - Dispersion medium content: 85 to 99 wt.% — description - Whole solute content: 1 to 15 wt.% — description - Carbon nanotube content: 50 to 90 wt.% of solute — description - Example 1: D10 1.93 μm — Example 1 - Example 1: D50 5.88 μm — Example 1 - Example 1: D90 15.6 μm — Example 1 - Example 1: viscosity 51 Pa·s — Example 1 - Example 1: dispersion efficiency 35.7 kWh/kg — Example 1 - Example 1: adhesion 28 gf/cm — Example 1 -

Example 1: DC-IR 1.252 ohm — Example 1 - Example 2: D10 2.26 μm — Example 2 - Example 2: D50 6.84 μm — Example 2 - Example 2: D90 18.2 μm — Example 2 - Example 2: viscosity 61 Pa·s — Example 2 - Example 2: dispersion efficiency 65.3 kWh/kg — Example 2 - Example 2: adhesion 26 gf/cm — Example 2 - Example 2: DC-IR 1.253 ohm — Example 2 - Example 3: D10 1.94 μm — Example 3 - Example 3: D50 6.77 μm — Example 3 - Example 3: D90 17.5 μm — Example 3 - Example 3: viscosity 47 Pa·s — Example 3 - Example 3: dispersion efficiency 34.3 kWh/kg — Example 3 - Example 3: adhesion 24 gf/cm — Example 3 - Example 3: DC-IR 1.248 ohm — Example 3 - Example 4: D10 2.18 μm — Example 4 - Example 4: D50 6.77 μm — Example 4 - Example 4: D90 17.1 μm — Example 4 - Example 4: viscosity 56 Pa·s — Example 4 - Example 4: dispersion efficiency 46.5 kWh/kg — Example 4 - Example 4: adhesion 27 gf/cm — Example 4 - Example 4: DC-IR 1.255 ohm — Example 4 - Example 5: D10 2.19 μm — Example 5 - Example 5: D50 5.81 μm — Example 5 - Example 5: D90 16.9 μm — Example 5 - Example 5: viscosity 34 Pa·s — Example 5 - Example 5: dispersion efficiency 31.2 kWh/kg — Example 5 - Example 5: adhesion 24 gf/cm — Example 5 - Example 5: DC-IR 1.250 ohm — Example 5 - Example 6: D10 2.01 μm — Example 6 - Example 6: D50 5.61 μm — Example 6 - Example 6: D90 16.3 μm — Example 6 - Example 6: viscosity 21 Pa·s — Example 6 - Example 6: dispersion efficiency 28.4 kWh/kg — Example 6 - Example 6: adhesion 23 gf/cm — Example 6 - Example 6: DC-IR 1.255 ohm — Example 6 - Example 7: D10 1.94 μm — Example 7 - Example 7: D50 5.87 μm — Example 7 - Example 7: D90 18.1 μm — Example 7 - Example 7: viscosity 36 Pa·s — Example 7 - Example 7: dispersion efficiency 33.1 kWh/kg — Example 7 - Example 7: adhesion 23 gf/cm — Example 7 - Example 7: DC-IR 1.261 ohm — Example 7 - Example 8: D10 2.22 μm — Example 8 - Example 8: D50 5.80 μm — Example 8 - Example 8: D90 17.6 μm — Example 8 - Example 8: viscosity 41 Pa·s — Example 8 - Example 8: dispersion efficiency 30.3 kWh/kg — Example 8 - Example 8: adhesion 27 gf/cm — Example 8 - Example 8: DC-IR 1.247 ohm — Example 8 - Example 9: D10 2.02 μm — Example 9 - Example 9: D50 6.09 μm — Example 9 - Example 9: D90 19.5 μm — Example 9 - Example 9: viscosity 73 Pa·s — Example 9 - Example 9: dispersion efficiency 68.9 kWh/kg — Example 9 - Example 9: adhesion 28 gf/cm — Example 9 - Example 9: DC-IR 1.244 ohm — Example 9

Relevant Experimental Evidence - Example 1: Bundle-type CNT dispersion was prepared with D50 5.88 μm and viscosity 51 Pa·s. - Example 2: A higher-viscosity formulation gave D50 6.84 μm and higher dispersion efficiency demand. - Example 5: Lower-viscosity partially hydrogenated NBR improved dispersion behavior. - Example 6: The lowest viscosity example showed D50 5.61 μm and 23 gf/cm adhesion. - Example 9: A high-viscosity formulation still met the CNT dispersion target but required more energy.

Technical Relevance This patent is relevant because it specifies partially hydrogenated NBR composition and residual unsaturation windows that are useful for formulation design. It demonstrates how HNBR-like materials can be tuned for dispersion and interfacial performance, which reflects the broader importance of hydrogenation control.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Direct hydrogenation patents (US7666950B2) emphasize reactor engineering and hydrogen pressure control, while simultaneous metathesis/hydrogenation patents (EP2473281B1, US9150669B2) combine chain scission and saturation in one step to lower Mw and Mooney viscosity.
- The low-MW HNBR patents (US7772328B2, US7951875B2) use metathesis first and then hydrogenation; they report product Mw around 54 kg/mol and Mooney viscosity as low as 3 in the best example, showing the benefit of precursor degradation before saturation.
- Catalyst systems differ substantially: US7666950B2 broadly allows Rh/Ru/Ti/Pt/Ir/Pd/Re/Os/Co/Cu catalysts, whereas EP2473281B1 and US9150669B2 focus on Ru/Os compounds for simultaneous metathesis/hydrogenation, and US8957161B2 uses salt-accelerated metathesis catalyst systems.
- Process conditions vary from batch hydrogenation pressures of 65-90 bar in the background of US7666950B2 to a claimed 3.5-100 bar window in the spinning-disk reactor process; the simultaneous metathesis/hydrogenation patents do not center on high-pressure batch operation but on catalyst efficiency and single-step conversion.
- Precursor composition control is explicit in US7666950B2 and EP2554558A1: NBR typically contains 40-90 wt.% diene and 10-60 wt.% nitrile, while the solvent-polymerization patent uses 70-99.9 vol% main solvent and 100°C polymerization for NBR synthesis.
- Post-treatment is addressed only in US7846995B2, which removes Ru residues to 10 ppm after 66 h at about 100°C using 0.5 g ion-exchange resin, highlighting downstream purification as a separate but important HNBR manufacturing step.
- The CNT-dispersion patents (US10686185B2, EP3348582B1) are not synthesis patents for HNBR itself, but they define useful partially hydrogenated NBR windows: RDB 0.5-40 wt.% and α,β -unsaturated nitrile content 20-50 wt.% for formulation performance.

4. CONCLUSION

The patent evidence supports three practical synthesis routes for hydrogenated NBR: direct hydrogenation of NBR in intensified reactors, metathesis-assisted molecular-weight reduction followed by hydrogenation, and simultaneous metathesis/hydrogenation in the presence of hydrogen. For conventional HNBR manufacture, the most defensible parameters are NBR feedstocks with about 40-90 wt.% diene and 10-60 wt.% nitrile, hydrogenation under Rh/Ru-based catalysis, and hydrogen pressures in the 3.5-100 bar range when using intensified reactor designs. For low-viscosity HNBR, the strongest evidence favors pre-metathesis or simultaneous metathesis/hydrogenation using Ru/Os catalyst systems, optional co-olefins, and conditions that drive Mw into the 20,000-250,000 range with Mooney viscosity below 50 and residual double bond content near the low end of the disclosed ranges. Downstream, ruthenium residue removal by functionalized ion-exchange resin is recommended when metathesis catalysts are used. Overall, the best-supported synthesis strategy is to tailor the NBR precursor first, then use catalyst-efficient hydrogenation or combined metathesis/hydrogenation to achieve low-RDB, low-Mooney HNBR.

5. REFERENCES

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- EP2554558A1 | Method for producing nitrile rubbers in organic solvents | Not disclosed | EP | 2013 | <https://patents.google.com/patent/EP2554558A1/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US7846995B2 | ... -containing catalyst residues from optionally hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7846995B2/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- US10686185B2 | Carbon nanotube dispersion liquid and manufacturing method thereof | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10686185B2/en>
- EP3348582B1 | Carbon nanotube dispersion liquid and manufacturing method thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3348582B1/en>